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# Analysis of EEG Signals in the DEAP Dataset for Emotion Recognition using Deep Learning Algortihms

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***Abstract*—Emotions are the behavioral responses representing mental state of a person. It is crucial to recognize the emotions of a person for human-computer interaction, to understand and respond to one's mental health. EEG signal provides a clear-sighted analysis of emotional state. It is a challenging task to recognize the patterns of multi channel EEG signals for emotion recognition using traditional Machine Learning approach. So, the Deep Learning algorithms are preferred more as they are successful in learning features and patterns to classify the the data. In this paper we have proposed Deep Learning models, CNN and LSTM Neural Networks to recognize the emotional states: Arousal, Valence and Dominance for the subjects 01-22 whose frontal videos were recorded and 23-32 whose frontal videos were non recorded in the DEAP dataset. CNN and LSTM Neural Networks are built with significant layers and the model is trained with required number of epochs and batch size after a number of trials to improve the accuracy of the emotion recognition. Finally, performance of the proposed models is evaluated with different evaluation metrics and the accuracy of both the Deep Learning models are compared for the subjects 01-22 and 23-32.**

***Keywords*-Emotion Recognition, EEG Signals, DEAP dataset, CNN (Convolutional Neural Network), LSTM (Long Short-Term Memory)**

## I. Introduction

Emotions are the person's response to a certain stimulus that represent their mental state. Individuals have varying emotional responses to the same situations. Since, wide range of emotions are exhibited by distinct people, emotion recognition is regarded as one of the most complex and intriguing research subjects in psychology. Emotion's multidimensional state represents human consciousness. Decision-making and Personal growth are influenced by thoughts, past experiences, or events that cause an emotion to be triggered. Negative emotions will impact on both psychological and physical health massively while Positive emotions can lead to higher living standards with healthy environment. In this day and age, negative emotions such as anxiety, loneliness, stress, hypertension, and other mental health concerns, affect an increasing number of people. As a result, Emotion Recognition plays a major role in improving health and mental state of the people. Emotion recognition is used in treatment of stress, fear and other mental health concerns by psychologists, moreover it also plays a crucial role in affective computing.

Emotion recognition can be carried out through facial expressions, speech recognition and physiological signals. Automatic facial analysis involves face recognition, facial features extraction and classification of expressions[1]. Though recognition of emotion through facial analysis are seems to be more superior than EEG signal analysis [2], the facial expressions can be faked most of the time. It is preferred to perform analysis of EEG signals for recognition of emotions [3]-[15]. EEG has a millisecond temporal resolution, making it suited for capturing the dynamics of emotional responses in real-time. Various databases are available for EEG analysis [4][5]. DEAP is a benchmark database for emotion recognition using physiological signals [5]. EEG is a popular neuroimaging technique that uses electrodes on the scalp to assess the electrical activity of the brain. Choosing more number of channels does not results in better accuracy nor choosing few frequency bands lower the accuracy of classifying the emotional states [6]. The interrelation between the EEG electrode selection and band pass frequency is necessary to understand [6]. EEG analysis is performed for classification of Arousal and Valence using machine learning models for 32 subjects [7]-[10].


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EEG analysis for emotion recognition has implications in a variety of domains, including psychology, neurology, human-computer interface, healthcare, and market research.

In this paper CNN and LSTM models are proposed for analysing EEG signals in the DEAP dataset for recognizing emotional states: Arousal, Valence, Dominance. This paper is organised as: Section I presents the introduction. Section II overviews on the DEAP dataset. Section III describes the proposed methodology. Section IV explains the experimental results. Section V concludes the aim of the paper followed by the references in the Section VI.

## II. DEAP DATASET

The DEAP dataset is a publicly available multimodal dataset for determining human emotional states. To our knowledge, this database has the most participants among freely available databases for analysing spontaneous emotions from physiological signals. The DEAP dataset can be downloaded from the official website: http://www.eecs.qmul.ac.uk/mmv/datasets/deap/index.html.
The EEG signals were recorded using a high resolution and a multi channel headset known as Emotiv Epoc. Experiments were carried out to create the database, in which 32 people were forced to watch 40 one-minute-long samples of selected videos. During each subject's 40 trials, various signals such as EEG, electromyograms, breathing region, plethysmographs, temperature, and so on were reported as 40 channel data, including signals from 32 channels of a conventional EEG headset and physiological signals from 8 channels. There are 32 files, one per each participant's recorded audio. Each trial has 63 signals, the first three of which are the baseline signals. Frontal videos were also recorded for the first 22 participants. After a series of filtering procedures, the data were down-sampled from 512 Hz to 128 Hz, and a band pass frequency filter with a frequency range of 4.0-45.0 Hz was used. EEG signals are divided into four bands: alpha (1-7 Hz), beta (8-13 Hz), theta (14-30 Hz), and gamma (30-45 Hz). Each movie was scored on a scale of 0 to 9 in terms of valence, arousal, dominance, and liking by the participants.

## III. PROPOSED METHODOLOGY

DEAP is a database for emotion recognition using various physiological signals. Among 40 channels used for recording of the physiological signals, the EEG signals were recorded in the first 32 channels using Emotiv Epoc. We have chosen a selective of 14 channels randomly out of 32 to get significant and accurate results. Initially the dataset was containing the pickled data which is retrieved using pickle load function. Signal windowing on the four band pass frequencies is used after band pass filtering. The window size chosen is small to ensure that more transient changes occurring in the signals are being captured. To ensure that the relevant information is not missed between consecutive segments the step size chosen is smaller. Smaller step size allows a higher degree of windows overlapping which allows continuous analysis of EEG signals.

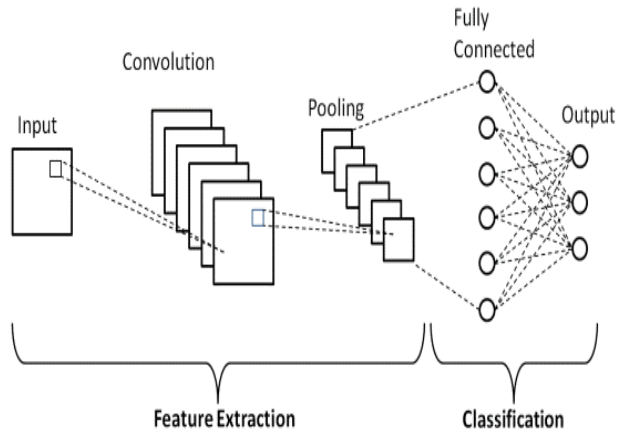


Fig. 1. Convolutional Neural Network

A FFT (Fast Fourier Transform) signal processing technique is applied to identify the relevant frequency components for emotional processing. By applying FFT, the model's accuracy and efficacy in the DEAP dataset can be increased. Different Neural Networks were built using CNN and LSTM with significant number of layers for the participants 01-22 and 23-32. The performances of these Deep Learning models are evaluated with confusion matrix, precision, recall and F1 score.

### A. CNN MODEL

The Convolution Neural Network has the capacity to find significant data in time series and image data. As a result, it is extremely useful for image-related tasks such as recognition of patterns, classifying of objects, and identification of images. CNNs can classify audio and signal data as well. The CNN is architecture is produced by different layers: fully-connected (FC), pooling, and convolutional, dropout layer and the activation function. CNN is applied in the various fields such as healthcare, automotive, social media.

Proposed Architectural Layers of CNN model:

- **Convolution 1D:** This layer performs the convolution operation on 1-dimensional input data. Convolution 1D detects the local patterns and spatial dependencies in the input data. The shape of the output from the convolutional layer is determined by the size and number of filters, as well as by the stride and padding settings.
- **MaxPooling 1D:** By selecting the maximum value inside each local region, the MaxPooling 1D layer reduces the spatial dimensions of the input. It functions independently on each input data channel. The reduction factor and the resulting output shape are determined by the size of the pooling window and the stride.
- **Flatten:** The convolutional layers and the fully connected layers are connected by the flatten layers in CNN. The multidimensional output of the previous layer is transformed into a one-dimensional vector by the flatten layer.


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The operation of flattening feds the output as input to the fully connected layers. It keeps the data's spatial structure in the convolutional layers while restructuring it into a format that the dense layers can process.

- **Batch Normalization:** The activations are normalised by subtracting the batch mean and dividing by the batch standard deviation. This normalisation step reduces the influence of shifting internal distributions during training, allowing the network to learn more efficiently. It can improve the network training speed and performance and also provide some regularisation benefits as well.
- **The Dropout** The input units are randomly set to 0 with a frequency of rate at each step during training time. The layer prevents overfitting.
- **Dense:** The layer learns non-linear relationships in the data and transforms the inputs into meaningful representations which is further used for prediction or classification.
- **Activation Functions:** The activation function decides whether a neuron should be activated or deactivated as it is necessary to prevent the non linearity. Activation functions include the ReLU, SoftMax, tanh, and Sigmoid functions.

### *B. LSTM MODEL*

Long Short-Term Memory (LSTM) is a form of Recurrent Neural Network (RNN) that handles sequential data such as time series, speech, and text. Because LSTM networks can learn long-term dependencies in sequential data, they are well suited for applications like language translation, speech recognition, and time series forecasting. Unlike traditional RNNs, which are constrained by the vanishing gradient problem, LSTMs can learn long-term associations by employing a technique known as gated recurrent units (GRUs). GRUs can utilise "forget" gates to decide which data from previous time steps should be discarded, and "update" gates to select how much data from the current time-step should be passed over to the next time-step.

Proposed Architectural Layers of LSTM model:

- **Bidirectional:** Bidirectional LSTM, for example, incorporates both past and future contextual information by keeping two sets of hidden states and cell states, one for forward and one for backward. This allows the network to capture dependencies in both directions, which is very valuable for jobs that involve sequential data.
- **Dropout:** At each training phase, Dropout sets a fraction of the input units to 0, essentially "dropping out" those units. This keeps the network from becoming overly reliant on particular input qualities or intricate interactions between them.
- **LSTM (Long Short-Term Memory):** The LSTM architecture is a form of recurrent neural network (RNN). Unlike standard RNNs, LSTM incorporates a memory cell and gating mechanisms to control information flow.
- **Dense:** Every neuron in a dense layer is coupled to every neuron in the previous layer. Each connection has a weight that is learned during the training phase. A dense layer's output is calculated by applying an activation function to the weighted sum of its inputs.
- **Activation:** By translating a neuron's output to a defined range or value, activation functions inject non-linearity into the neural network. They are applied to the output of a layer or a neuron element by element.

## IV. RESULTS

The CNN and LSTM models performs better when compared to traditional machine learning approaches. It takes approximately 3 hours to train the model using LSTM for the subjects 01-22 while using CNN it takes an hour. For the subjects 23-32 in the DEAP dataset, about 30 minutes of time is taken for training the model using LSTM while using CNN it takes about 5 minutes. On comparision of both models, though LSTM takes longer time to train the models in both the cases but it results in a better and a higher accuracy of emotional states than CNN. LSTM is effective and gives enahanced result for about 10% more than CNN. The accuracy curves are plotted for both the models to ensure there is no overfitting or underfitting of the model.

We observed that LSTM outperforms the CNN in both the cases. The accuracy of 62.70% for Arousal, 62.97% for Valence and 64.14% for Dominance is obtained using CNN while using LSTM the accuracy is 82.40%, 78.28%, 83.36% for Arousal, Valence and Dominance respectively for the subjects 01-22. The accuracy of 33.35% for Arousal, 32.93% for Valence and 33.25% for Dominance is obtained using CNN while using LSTM the accuracy is 63.15%, 62.06%, 59.85% for Arousal, Valence and Dominance respectively for the subjects 23-32. The performance of the both the models are evaluated using performance metrics.

### *A. Accuracy and Performance Metrics for the subjects 01 – 22 using CNN*

Table I serves the performance evaluations carried on the proposed Deep Learning model. Figure 2 and 3 represents the training testing accuracy and the performance evaluation through confusion matrix for emotional state Valence. Similarly, the training and testing curve is obtained and the performance has been evaluated for Arousal and Dominance as shown in Fig 4.

TABLE I
PERFORMANCE EVALUATION USING CNN FOR THE SUBJECTS 01-22

| Evaluation Metrics | Arousal | Valence | Dominance |
|---|---|---|---|
| Accuracy | 62.70% | 62.97% | 64.14% |
| Precision | 64.26% | 64.53% | 65.31% |
| Recall | 62.70% | 62.97% | 64.14% |
| F1 Score | 62.80% | 63.31% | 64.25% |


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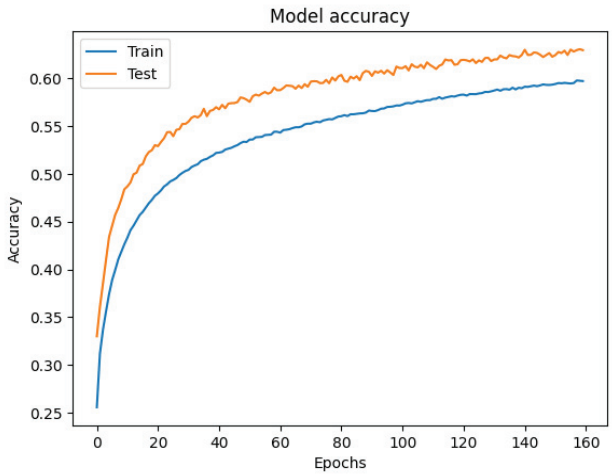


Fig. 2. Training and Testing Accuracy of Valence for subjects 01-22

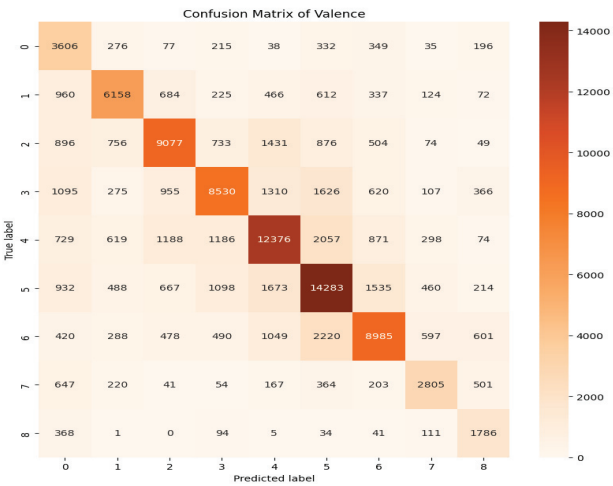


Fig. 3. Confusion Matrix of Valence for subjects 01-22

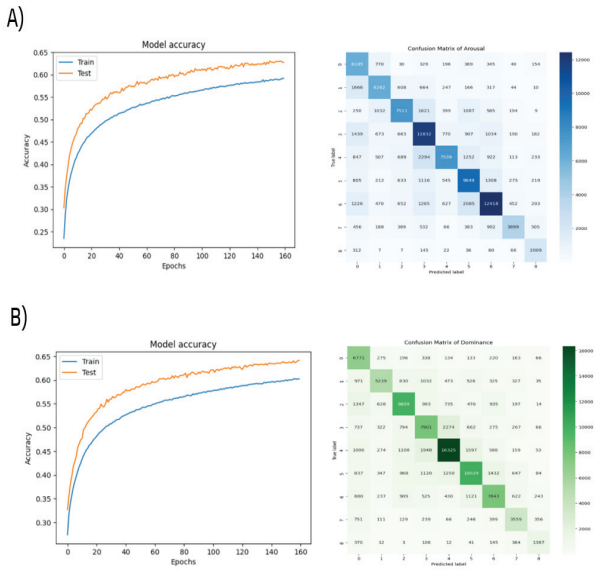


Fig. 4. A) Arousal and B) Dominance of subjects 01-22

## B. *Accuracy and Performance Metrics for the subjects 23 – 32 using CNN*

Table II shows the performance metrics carried on the proposed Deep Learning model. Figure 5 and 6 displays the training testing accuracy and the performance evaluation through confusion matrix for emotional state Valence. Similarly, the training and testing curve is obtained and the performance has been evaluated for Arousal and Dominance as shown in Fig 7.

TABLE II
PERFORMANCE EVALUATION USING CNN FOR THE SUBJECTS 23-32

| Evaluation Metrics | Arousal | Valence | Dominance |
|---|---|---|---|
| Accuracy | 33.35% | 32.93% | 33.25% |
| Precision | 31.06% | 26.98% | 27.25% |
| Recall | 33.35% | 32.93% | 33.25% |
| F1 Score | 28.73% | 25.62% | 27.25% |

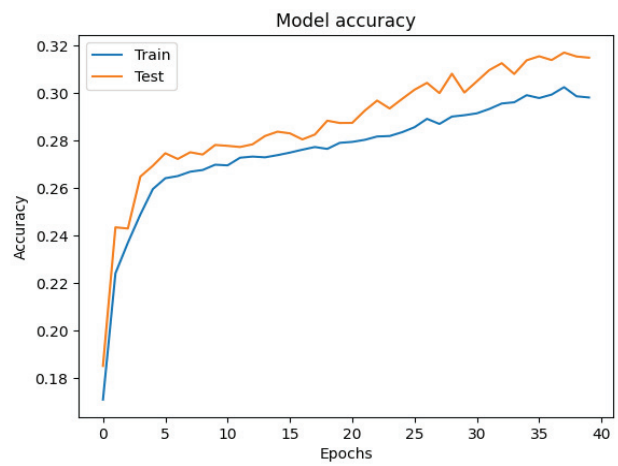


Fig. 5. Training and Testing Accuracy of Valence for the subjects 23-32

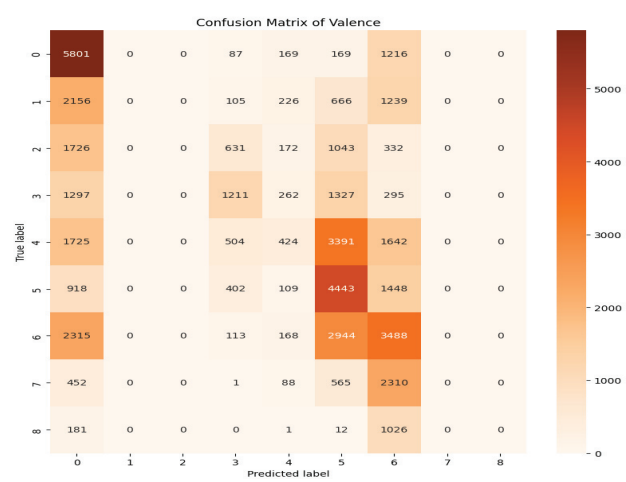


Fig. 6. Confusion Matrix of Valence for the subjects 23-32



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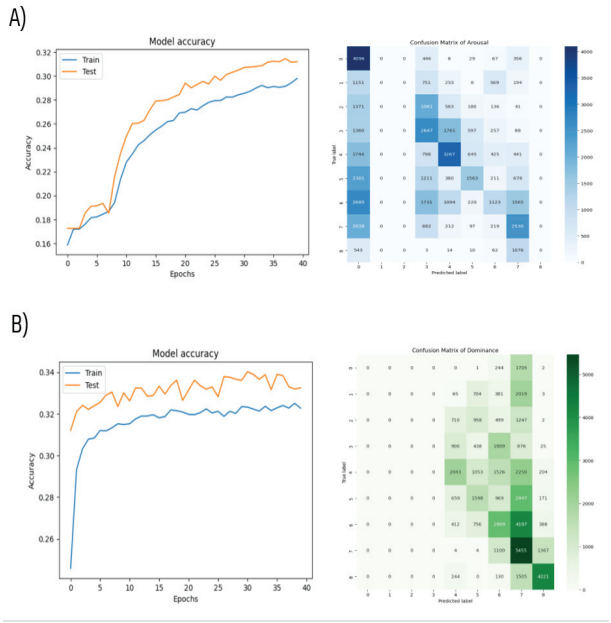


Fig. 7. A) Arousal and B) Dominance for the subjects 23-32

### C. Accuracy and Performance Evaluation for the subjects 01 – 22 using LSTM

Table III depicts the performance evaluation carried out on the proposed Deep Learning model. Figure 8 and 9 represents the training testing accuracy and the performance evaluation through confusion matrix for emotional state Arousal. Similarly, the training and testing curve is obtained and the performance has been evaluated for Valence and Dominance as shown in Fig 10.

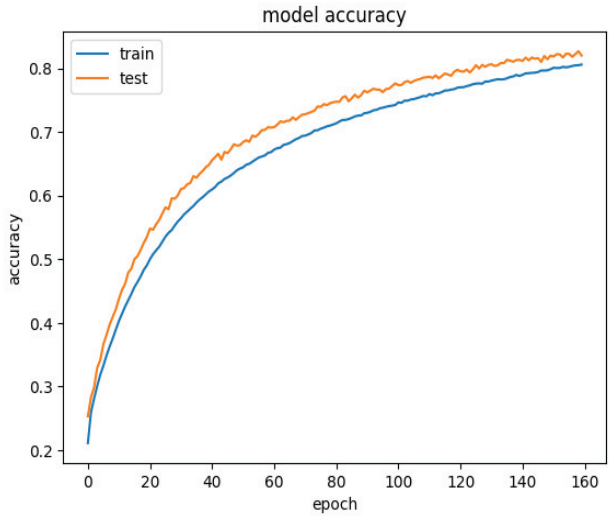


Fig. 8. Training and Testing Accuracy of Arousal for the subjects 01-22

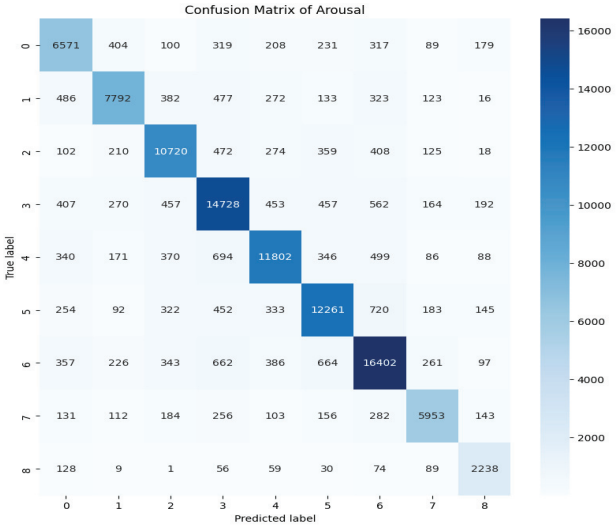


Fig. 9. Confusion Matrix of Arousal for the subjects 01-22

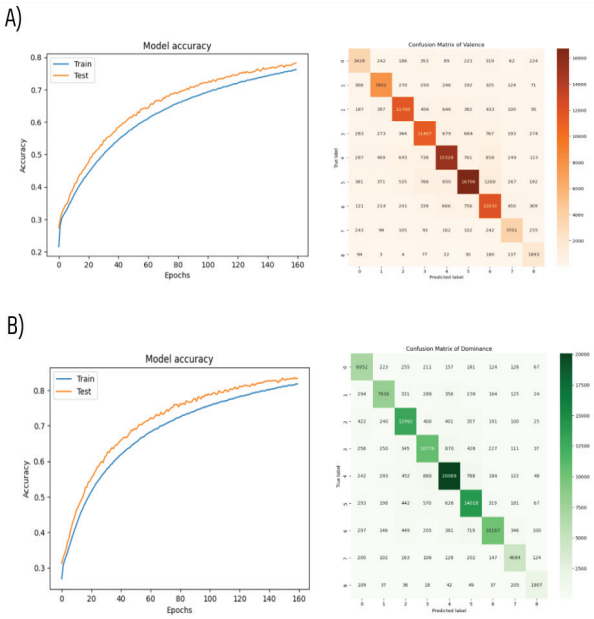


Fig. 10. A) Valence and B) Dominance of subjects 01-22

TABLE III
PERFORMANCE EVALUATION USING LSTM FOR THE SUBJECTS 01-22

| Evaluation Metrics | Arousal | Valence | Dominance |
|---|---|---|---|
| Accuracy | 82.40% | 78.28% | 83.36% |
| Precision | 82.49% | 78.73% | 83.48% |
| Recall | 82.40% | 78.28% | 83.36% |
| F1 Score | 82.41% | 78.41% | 83.38% |


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### D. Accuracy and Performance Evaluation for the subjects 23 – 32 using LSTM

Table IV represents the performance evaluation carried out on the proposed Deep Learning model. Figure 11 and 12 depicts the training testing accuracy and the performance evaluation through confusion matrix for emotional state Arousal. Similarly, the training and testing curve is obtained and the performance has been evaluated for Valence and Dominance as shown in Fig 13.

TABLE IV
PERFORMANCE EVALUATION USING LSTM FOR THE SUJECTS 23-32

| Evaluation Metrics | Arousal | Valence | Dominance |
|---|---|---|---|
| Accuracy | 63.15% | 62.06% | 59.85% |
| Precision | 64.44% | 62.62% | 60.82% |
| Recall | 63.15% | 62.06% | 59.85% |
| F1 Score | 63.32% | 62.06% | 59.93% |

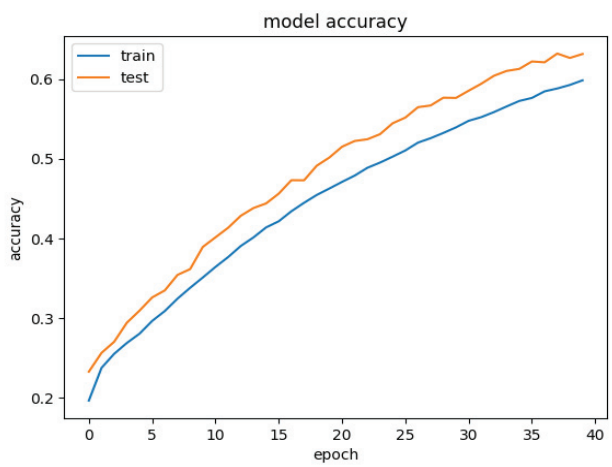


Fig. 11. Training and Testing Accuracy of Arousal for the subjects 23-32

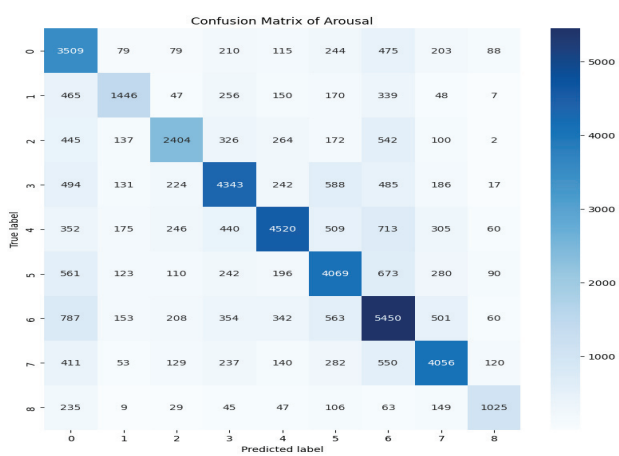


Fig. 12. Confusion Matrix of Arousal for the subjects 23-32

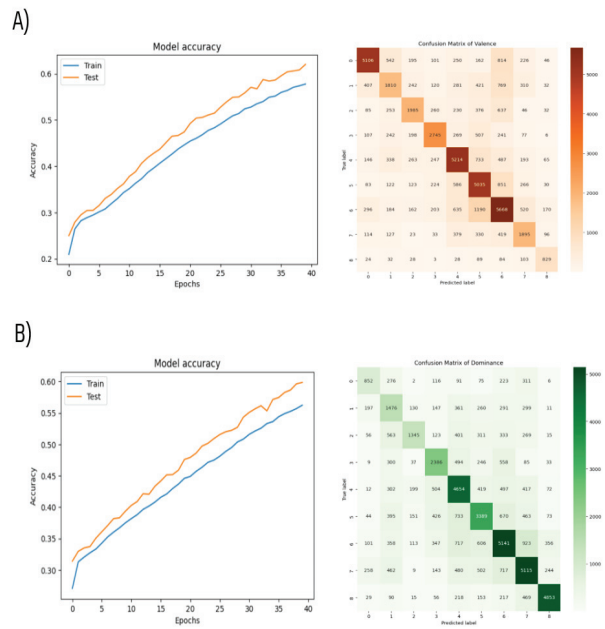


Fig. 13. A) Valence and B) Dominance for the subjects 23-32

## V. CONCLUSION

We have proposed CNN and LSTM model for analysing EEG signals of the participants for emotion recognition. On analyzing the results, we observed the accuracy of the emotional states: Arousal, Valence, Dominance is higher for the participants whose frontal videos has also been recorded simultaneously along with their SAM ratings i.e., for the subjects 01 – 22 than the participants whose frontal videos were not recorded i.e for the subjects 23 – 32 in the DEAP dataset. This infer that a large dataset improves the accuracy of the deep learning model by learning the features and patterns accurately. On comparison of CNN and LSTM algorithms, It concludes, LSTM algorithm outperforms by enhancing the accuracy 10% higher than the CNN model. Table V shows the testing accuracy report of the emotional states: Arousal, Valence, Dominance for both CNN and LSTM models.

TABLE V
TESTING ACCURACY REPORT

| Subjects | Classifier | Arousal | Valence | Dominance |
|---|---|---|---|---|
| 01-22 | CNN | 62.70% | 62.97% | 64.14% |
| 01-22 | LSTM | 82.40% | 78.28% | 83.36% |
| 23-32 | CNN | 33.35% | 32.93% | 33.25% |
| 23-32 | LSTM | 63.15% | 62.06% | 59.85% |

In the future work, Deep Learning models can be built for the dataset containing frontal videos of the participants in the DEAP and cross validate the results with the results obtained in this paper for the SAM ratings given by the same


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participants whose frontal videos are recorded. In addition, can work upon different algorithms other than CNN and LSTM to improve accuracies for the recognition of emotional states in DEAP dataset. Two different deep learning models can be combined to form a Hybrid Neural Network to improve the accuracy.

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